

Anthropogenic Surface Disturbance Mapping: Temporal Mapping (Simplified Approach)

The BEACONS Project

Updated: August 18, 2026

Overview

The purpose of this document is to provide a summary of a simplified approach to generate temporal maps of anthropogenic surface disturbances, both linear and areal (polygonal), across areas of interest (AOIs) spanning the Yukon Territory's northwest boreal regions. A more resource intensive approach to temporal mapping is detailed in our procedures manual and case study. Datasets used to develop this method were obtained from previous work conducted by the BEACONS project and were assigned disturbance dates based on their respective years of appearance using the Google Earth Pro 'Timelapse' tool. All GIS work was conducted in QGIS (open-source) to increase both the accessibility and applicability of the temporal mapping procedures outlined below.

This document is divided into two sections:

- Aging Surface Disturbances:
 - Detailed outline of the procedures used to digitize and date the appearance and expansion of anthropogenic surface disturbances through time.
- Project Tables:
 - Templates illustrating the format of the tables used to track work progress and relevant attribute information using Google Sheets & QGIS.

Aging surface disturbances

Surface disturbance layers containing existing linear and polygonal features (e.g., sd_line & sd_poly), should never be modified.

Project Setup

This section focuses on the temporal aspects of the project. For additional details of setting up a disturbance mapping project from scratch, please see the General Procedures manual.

Add Temporal Data Field(s):

Linear and polygonal surface disturbances may be dated with i) one field to record the year of disturbance (DIST_YR) or ii) using a four-category system developed to ensure any degree of uncertainty surrounding the appearance of a disturbance is adequately captured in the dataset. The confidence level associated with each field can then be considered in analysis. The four-category system is as follows:

- DIST_YR: Date of occurrence for a given disturbance identified to the year.
 - Usage: High temporal resolution allows for a disturbance to be dated to the year of imagery in which it first appears, with no signs of the disturbance present in the imagery for the previous year.
- YR_VISIBLE: Year of imagery when disturbance first becomes clearly identifiable:
 - Usage: When a degree of uncertainty exists surrounding the appearance of a disturbance on the landscape (e.g., poor resolution imagery in buffering years), the feature is dated to the earliest year in which it becomes clearly identifiable in the imagery.
- YR_IMAGERY: Year of imagery used to digitize the feature.
 - Usage: In instances where “YR_DSTRBNC” & “YR_VISIBLE” fields are blank, this field is referred to as the earliest known date of disturbance. In most instances, disturbance is assumed to predate this.
- YR_EST: Estimated year of disturbance & the value used to map features.
 - Usage: Based on information gathered, best estimate of the date of occurrence for a given disturbance on the landscape.

Optionally, another field (YR_RPLCED) may be used to record the year a disturbance disappears from the landscape, either from vegetation recovery or replacement by another (typically larger) disturbance feature.

Manual Dating Procedure

Age Surfaces Disturbances:

This workflow utilizes Google Earth Pro and Google Timelapse to assign dates to disturbance features.

Method 1: Navigate to the AOI and Scroll through the available years in Google Timelapse and Google Earth Pro to observe changes to the landscape.

- Once the feature appears, return to the QGIS project to assign a date to the DISTURBANCE_YEAR attribute field.

Method 2: Import project shapefiles into Google Earth Pro, enabling users to digitize the timelapse imagery. The workflow for this process is as follows:

1. Import disturbance shapefile into Google Earth Pro (.klm file) and digitize features as needed.
2. Convert back to shapefile using ESRI toolbox or QGIS equivalent. Attributes must then be manually entered into QGIS.

Digitize Historic Features:

Additional digitizing may be required in instances where:

- Newly identified disturbance features not included as digitized features within the current datasets, including features in proximity to previously existing (digitized) features, or in newly disturbed areas.
- Existing features that have changed in shape and size since they were last digitized. In these cases, the entire feature that has changed is re-digitized, not just the additions.
- Existing features that do not appear to represent the feature correctly e.g., errors or features that were previously digitized using older/coarser imagery.

Substitutions and Verification:

- **Utilize the FLAG and NOTE fields to notify reviewers of features that require additional verification. See the procedures manual for further details.**
- When significant temporal gaps exist in the available imagery for a particular AOI, government records (e.g., YESAB registry, Yukon Water Board, and GeoYukon) can be used to assist in identifying the appearance of linear and polygonal surface disturbances.
- Local knowledge holders can also provide information on disturbance timelines, although this is a more labour-intensive approach.

Disturbance attributes

Table 1 is similar to the attribute table presented in the General Procedures manual but also includes the extra attributes relevant to temporal mapping.

Table 1. List of attributes included in the areal and linear disturbance shapefiles.

Attribute	Data Type	Domains	Description
REF_ID	String (254)		Unique feature reference ID - leave this blank.
DATABASE	String (254)		Sub-database to which the feature belongs.

Attribute	Data Type	Domains	Description
TYPE_INDUSTRY	Text (20)	Appendix 1	Major classification of disturbance feature by industry
TYPE_DISTURBANCE	Text (30)	Appendix 1	Sub classification of disturbance feature
CREATED_BY	Text (25)	USER	User that aged the disturbance
CREATED_DATE	Date		Date feature was created/digitized
FLAG	Short Integer (1)		Feature is a high priority for validation (mark with value 1)
NGHBR_EST	Short Integer (1)		Feature is assigned age estimate based on the date of appearance of an associated feature (mark with a value of 1)
NOTES	Text (200)		Notes
DIST_YR	Integer (4)		Date of occurrence for a given disturbance identified to the year; to be used if only one field for ageing disturbances is to be used, without recording the degree of uncertainty associated with a disturbance date.
YR_DSTRBNC	Integer (4)		Date of occurrence for a given disturbance identified to the year (high confidence).
YR_VISIBLE	Integer (4)		Year of imagery when disturbance first becomes clearly identifiable, but degree of uncertainty exists (e.g., poor resolution imagery in buffering years) (medium confidence).
YR_IMAGERY	Integer (4)		Year of imagery used to digitize the feature. In instances where "YR_DSTRBNC" & "YR_VISIBLE" fields are blank, this field is referred to as the earliest known date of disturbance. In most instances, disturbance is assumed to predate this.
YR_EST	Integer (4)		Based on information gathered, best estimate of the date of occurrence for a given disturbance (low confidence).

Attribute	Data Type	Domains	Description
YR_RPLCED	Integer (4)		Year a surface disturbance was replaced by vegetation or another disturbance (optional).